

YMNOTES MEDICAL · LECTURE EXTRAS

Complications of Peptic Ulcer

Supplement to Digestive System — Version 3

Chapter 03 · The Stomach and Duodenum · Surgery

Only what the lecture added beyond the book — arranged by book page.

Book pages 34 – 57 LECTURE EXTRAS

Clear structure. Connected understanding.



GUIDE Map

Where each extra belongs in the book

Teach the book page first, then the extras listed against it.

Everything below was taught in the lecture but is **not** in the book. Content the book already covers (the list of complications, the general preparation for surgery, the D.D., G.O.O. work-up) is left out. Tap a row to jump to it.

BOOK PAGE	BOOK SECTION IT ATTACHES TO	EXTRA FROM THE LECTURE	
p. 34	<i>Anatomy and Physiology — Gastric Anatomy</i>	Sympathetic supply of the stomach	3 →
p. 42–43	<i>Peptic Ulcer Disease (definition, sites)</i>	Acute vs chronic peptic ulcer · Chronic gastric ulcer types I–IV	3 →
p. 53	<i>Complications of peptic ulcer</i>	Indications of surgery in chronic DU · The three problem-solving keys	4 →
p. 54	<i>I-Perforated peptic ulcer — Pathology & Stages</i>	Incidence & mortality · Factors with increased mortality · Diurnal peak & fasting · Transmural necrosis · Triphasic peritoneal response → SIRS	4 →
p. 55–56	<i>How to investigate? · Figure 3.8</i>	Plain X-ray: lower chest + abdomen · What U/S & CT can and cannot show · Normal vs perforation chest X-ray	6 →
p. 56	<i>Surgical exploration · Omental patch (Figure 3.9)</i>	Lap. vs open laparotomy · Omental patch closure in 3 steps	7 →
p. 57	<i>Gastric ulcer suspicious for malignancy</i>	Partial gastrectomy · Billroth I vs II EXAM	8 →

How the lecture framed every complication

Items for each topic

1. Pathophysiology — *why did the perforation happen, and what happens after it?*
2. How to reach the diagnosis?
3. How to manage?
4. Operative notes

Four questions to answer

1. How to reach a clinical diagnosis? What are the expected complications?
2. What is the D.D.?
3. How to reach the definite diagnosis? (Investigations)
4. How to treat?

 Lecturer's spoken / handwritten note

 Contrast with the book

EXAM  Flagged for the exam in the lecture



BOOK PAGE
34

Anatomy — nerve supply

Add after *Gastric Anatomy* (the book gives regions and sphincters, no nerve supply)

Sympathetic supply of the stomach NEW

- ◆ From the **greater splanchnic nerve** → **celiac ganglia** → stomach.
- ◆ It carries **visceral pain sensation**.
- ◆ Action: **pyloric contraction (spasm)**.

BOOK PAGE
42–43

Peptic Ulcer Disease — types and sites

Add after “GUs are most commonly seen on the lesser curve near the incisura...”

Types of peptic ulcers: acute vs chronic NEW

I. Acute ulcer

Depth Mucosa & submucosa **only**
Healing By **epithelialization**
Usual problem **Bleeding** (usually)

II. Chronic ulcer

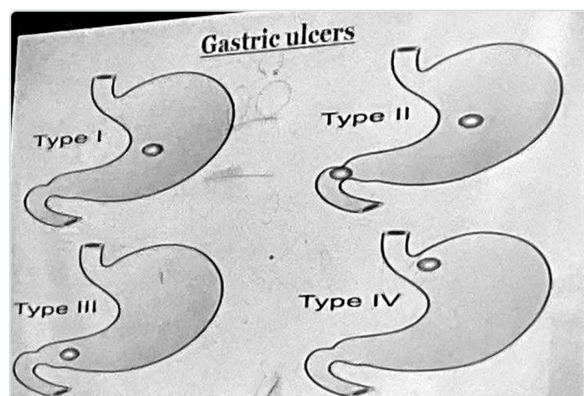
Depth Mucosa & submucosa + **muscle layer**
Healing By **fibrosis** (scar)
Usual problem The scar **causes obstruction**

LECTURE QUESTION · “WHAT IS THE MEANING?”

Chronic → healing by fibrosis (scar) → obstruction (see pyloric stenosis, book p. 57).

Chronic gastric ulcer (GU) — sites NEW

Type	Site	Acid secretion
Type I	Usually at the lesser curvature (ulcer-bearing area)	Low or normal
Type II	Body of the stomach , associated with a duodenal ulcer (2 ulcers: stomach + duodenum)	Excessive
Type III	Pre-pyloric ulcer — behaves like a DU	Excessive
Type IV	Near the GE junction	Normal or low



Lecture figure — Gastric ulcers, types I–IV. Classification depends on the **site** and the **acid secretion**.



BOOK PAGE

53

Complications of peptic ulcer

The complication list itself is in the book (the lecture slide was marked "in the book"). Add these after it.

Indications of surgery in chronic duodenal ulcer (DU) NEW

1. Ulcer **fails to heal** — severe symptoms.
2. **Tendency** of complications.
3. **Complicated** cases.

Keys: problem solving NEW

The lecture treats the surgical complications as three problems to solve:

1. **Perforated** peptic ulcer
2. **Bleeding** peptic ulcer
3. **G.O.O.** (gastric outlet obstruction) → pyloric stenosis: **cicatricial vs malignancy**.

BOOK PAGE

54

I-Perforated peptic ulcer

Add to *Pathology* and *Stages*

LECTURE NOTE · WHAT "PERFORATION" MEANS

The **serosa** gives way and the contents leak into the **peritoneal cavity**. It has an **acute onset and a rapid course**.

Incidence and mortality NEW

- ◆ Incidence: **2%–10%** (of peptic ulcers) progress to perforation.
- ◆ Mortality rate: between **8% and 15%**.

Factors with increased mortality NEW

- 1 **Preoperative shock** (systolic BP < 90 mmHg)
- 2 **Severe medical comorbidity** — e.g. liver failure, kidney failure
- 3 **Presentation delay** > 24–48 hours
- 4 **Defect size** > 2 cm

A diurnal peak of ulcer perforations NEW

1. More perforations occur in the **morning**.
2. Perforation risk is increased by **fasting**, such as during **Ramadan**.
(due to variation in acid release and exposure)

LECTURE NOTE

Fasting → the stomach is empty → the HCl is **concentrated**.



BOOK PAGE

54

Perforated peptic ulcer — pathophysiology (continued)

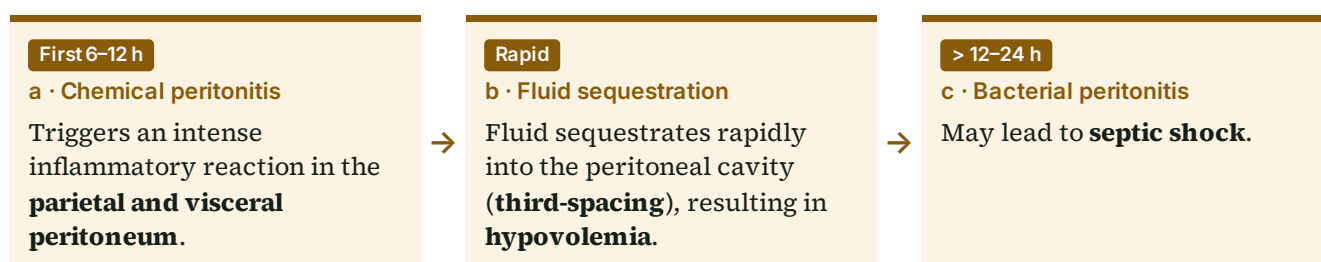
The book has the three stages (chemical irritation, illusion, septic peritonitis). The lecture adds timing, mechanism and the systemic end-point.

1. Mucosal erosion to full-thickness necrosis NEW

Chronic ulcer patient + exposure to an ulcerogenic agent → **progressive transmural necrosis** through the:



2. Triphasic peritoneal response NEW



LECTURE NOTES

Chemical phase: the HCl essentially **burns the parietal peritoneum**.

Fluid phase (“peritoneal reaction”): the peritoneum secretes alkaline fluid to neutralise the HCl — this is the fluid that is lost into the third space.

3. Systemic Inflammatory Response & Sepsis NEW

Septic peritonitis → **SIRS** → may progress to **multi-organ failure**.

BOOK ↔ LECTURE

Book stage (p. 54)	Lecture phase	Lecture adds
a. Chemical irritation	Chemical peritonitis	First 6–12 h; parietal + visceral peritoneum
b. Stage of illusion	Fluid sequestration	Third-spacing → hypovolemia
c. Septic peritonitis	Bacterial peritonitis	> 12–24 h; then SIRS & sepsis → multi-organ failure



BOOK PAGE
55–56

How to investigate? — radiology

Add to *A. Radiological* and next to *Figure 3.8 — Air under diaphragm*

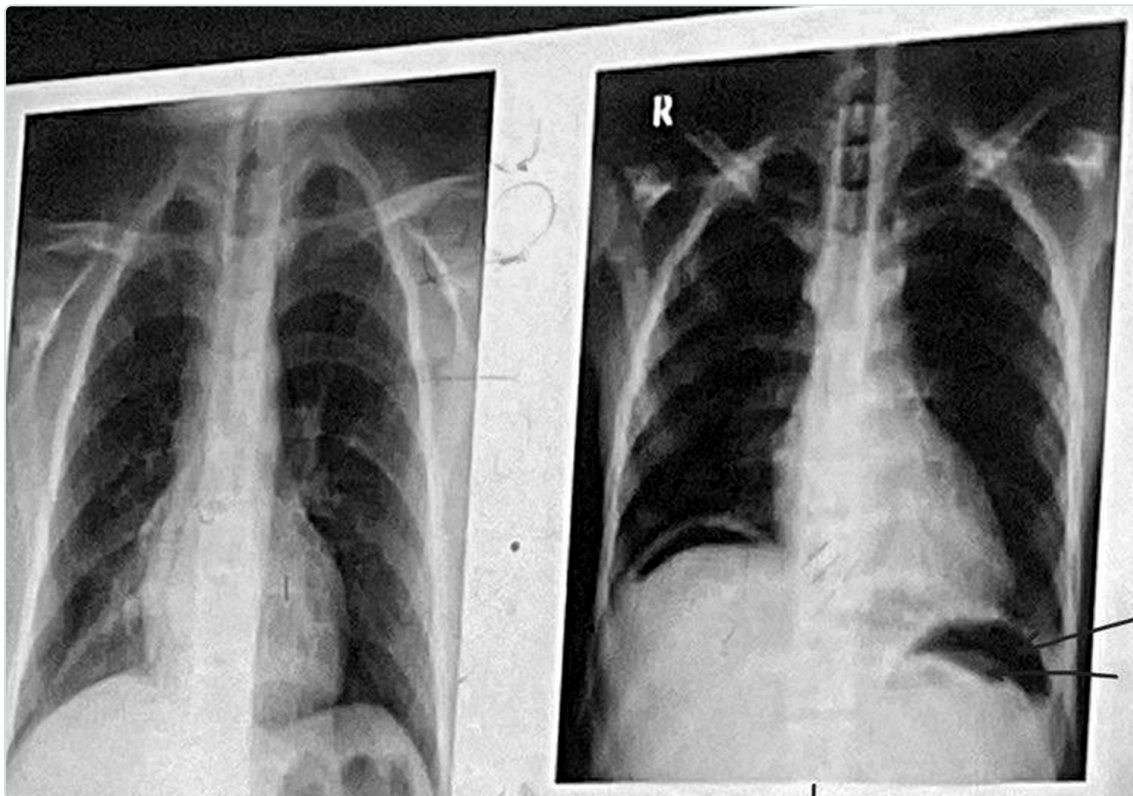
Plain X-ray — what to request NEW

- ◆ Request **lower chest + abdomen** (the book says “abdomen”).
- ◆ Taken **erect** — the patient must be standing.

U/S and CT — what they show NEW

- ◆ They show the **peritoneal fluid** that collects as a result of the perforation.
- ◆ They do **not** show the perforation (the hole) itself.

Chest X-ray: normal vs perforation NEW



Lecture figure — Left: **normal**. Right: **perforation** (arrows: lecturer's labels).

LECTURE NOTE · THE TWO ARROWS ON THE PERFORATION FILM

Upper arrow: **air under the diaphragm**. Lower arrow: **air in the stomach**. The lecturer labelled both so the two air shadows are not confused.



BOOK PAGE
56

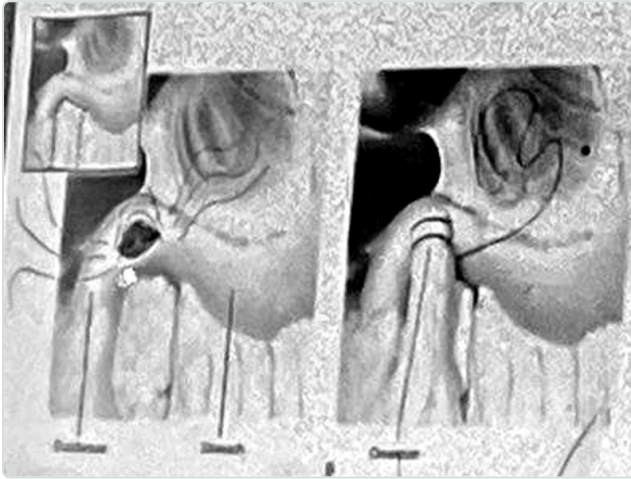
Management — operative notes

Add to *Surgical exploration of the abdominal cavity* and *Figure 3.9*

Surgery is an emergency NEW

- ◆ Approach: **laparoscopic vs open laparotomy** (the book mentions only a midline incision).
- ◆ We do abdominal exploration → **omental patch closure + peritoneal irrigation & drains**.

Omental patch closure — the three steps NEW



Lecture figure — sutures placed (left), tied over the omentum (right).

1. **Interrupted full-thickness sutures** are placed across the perforation site **without tying**.
2. A **vascularized pedicled plug of greater omentum** is laid over the defect.
3. The sutures are tied **over the omentum** to seal the perforation.



BOOK PAGE
57

Suspicious gastric ulcer — partial gastrectomy

Add after “In case gastric ulcer in suspicious case for malignancy gastrectomy can be done.”

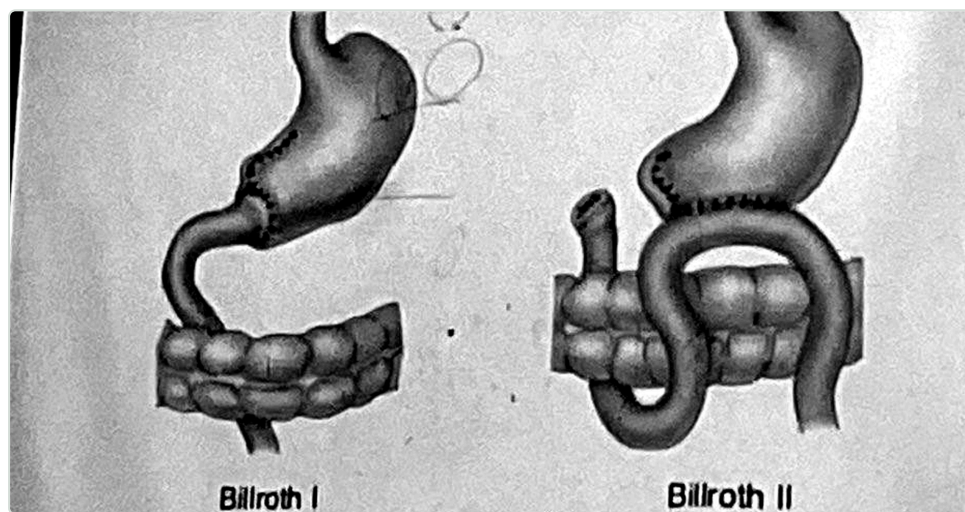
Perforated gastric ulcer suspicious for malignancy NEW

- ◆ In a suspicious case for **malignancy** → **partial** gastrectomy can be done (the book says “gastrectomy”).

LECTURE QUESTION

“Omental patch???” — the slide left this open for discussion: is an omental patch enough for a perforated gastric ulcer that is suspicious for malignancy?

Reconstruction after partial gastrectomy: Billroth I vs II EXAM NEW



Lecture figure — Billroth I (left) and Billroth II (right).

	Billroth I	Billroth II
Anastomosis	Gastro-duodenostomy	Gastro-jejunostomy
Meaning	The stomach is joined to the duodenum	The stomach is joined to the jejunum

CROSS-REFERENCE

Billroth II appears once more in the book, on **p. 66** (lower radical gastrectomy for cancer of the lower 1/3 + gastro-jejunostomy). Billroth I is not in the book.

رَبِّ زِدْنِي عِلْمًا

[سورة طه: 114]



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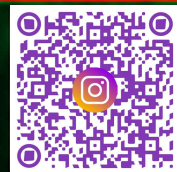
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